PES University, Bengaluru

(Established under Karnataka Act 16 of 2013)

END SEMESTER ASSESSMENT (ESA) - DEC 2023

UE21CS352A - Machine Intelligence

Total Marks : 100.0

1.a. Write the PEAS framework to design an “Intelligent system for an AI-powered personal assistant”. Write the name of each component in PEAS. Provide specific details for each component of the PEAS framework based on the context of the “personal assistant” scenario. (5.0 Marks)

1.b. Suppose you are evaluating the performance of a sentiment analysis model for customer reviews. The confusion matrix for the model's predictions on a test set is given below:

	Predicted Positive	Predicted Negative
Actual Positive	350	50
Actual Negative	40	860

Calculate the following performance metrics:

a) Accuracy

b) Precision for the positive class

c) Recall for the negative class

d) F1 Score (4.0 Marks)

1.c. What do you mean by Bias Variance Decomposition? **3M**

Recall the logistic activation function σ and the tanh activation function:

$$\sigma(z) = \frac{1}{1+e^{-z}}$$

$$\tanh(z) = \frac{e^{z} - e^{-z}}{e^{z} + e^{-z}}$$

1. One of the difficulties with the logistic activation function is that of saturated units. Briefly explain the problem, and whether switching to tanh fixes the problem. **1M**
2. Briefly explain one way in which using tanh instead of logistic activations makes optimization easier **1M**

(5.0 Marks)

1.d. Consider the below dataset

No	Feature 1	Feature2	Feature3	Class Label
1	T	T	1.0	Positive
2	T	T	6.0	Positive
3	T	F	5.0	Negative
4	F	F	4.0	Positive
5	F	T	7.0	Negative
6	F	T	3.0	Negative
7	F	F	8.0	Negative
8	T	F	7.0	Positive
9	F	T	5.0	Negative

(a) What is the entropy of this collection of training examples with respect to the positive class?
(b) What are the information gains of feature1 and feature2 relative to these training examples? (5.0 Marks)

1.e. # Import necessary libraries

```
from sklearn.datasets import load_iris
from sklearn.model_selection import train_test_split
from sklearn.tree import DecisionTreeClassifier
from sklearn.metrics import accuracy_score, classification_report

# Load the Iris dataset
iris = load_iris()
X = iris.data
y = iris.target

# Split the dataset into training and testing sets
X_train, X_test, y_train, y_test = train_test_split(X, y, test_size=0.2,
random_state=42)

# Initialize the Decision Tree Classifier
dt_classifier = DecisionTreeClassifier(random_state=42)

# Train the Decision Tree model
dt_classifier.fit(X_train, y_train)

# Make predictions on the test set
y_pred = dt_classifier.predict(X_test)

# Evaluate the model
accuracy = accuracy_score(y_test, y_pred)
report = classification_report(y_test, y_pred)

print(f"Accuracy: {accuracy:.4f}\n")
print("Classification Report:\n", report)
```

You are provided with a Python code snippet that utilizes a Decision Tree Classifier to classify the Iris dataset.

i) Briefly explain the purpose and functionality of each major section of the code.

ii) Discuss the significance of using the Iris dataset and why it is suitable for demonstrating the capabilities of a Decision Tree Classifier.

iii) Explain the key hyperparameters used in the DecisionTreeClassifier initialization and their roles in controlling the behavior of the decision tree.

iv) Analyze the evaluation metrics presented in the code (accuracy and classification report) and interpret their implications for the model's performance. (6.0 Marks)

2.a.

Derive the gradient descent training rule assuming that the target function representation is:

$$o_d = w_0 + w_1 x_1 + \ldots + w_n x_n.$$

Define explicitly the cost/error function E, assuming that a set of training examples D is provided, where each training example $d \in D$ is associated with the target output t_d.

Prove that the LMS training rule performs a gradient descent to minimize the cost/error function E.

(8.0 Marks)

2.b. Consider a Convolutional Neural Network (CNN) with a single convolutional layer followed by a max-pooling layer. The input image has dimensions of 32x32 pixels with three color channels (RGB). The convolutional layer uses 16 filters of size 3x3 with a stride of 1. The max-pooling layer uses a 2x2 pooling window with a stride of 2.

i) Calculate the number of parameters in the convolutional layer.

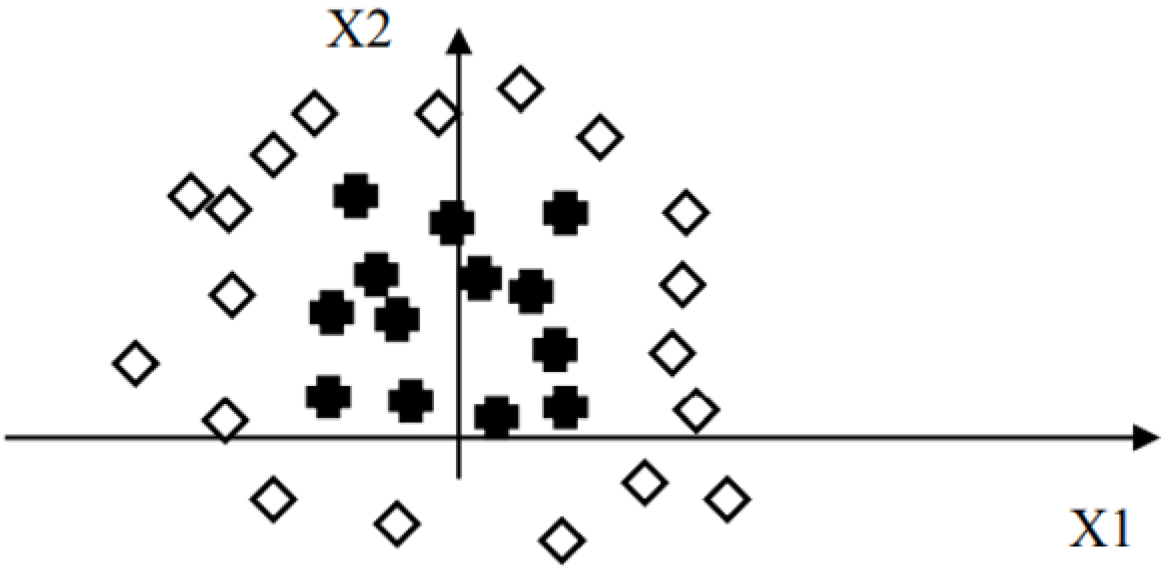
ii) Determine the dimensions of the output feature map after the convolutional layer.

iii) Calculate the number of parameters in the max-pooling layer.

(8.0 Marks)

2.c. Explain the significance of optimizers in the training process of neural networks. Provide a concise overview of two popular optimizers, such as RMSprop and SGD (Stochastic Gradient Descent). Highlight a key difference between these optimizers and discuss how their characteristics contribute to more efficient model training. (5.0 Marks)

2.d. Suppose that we want to build a neural network classifier that classifies two-dimensional data (i.e., X = [x1, x2]) inti two classes: diamonds and crosses. We have a set of training data that is plotted as follows:



Draw or Describe the structure of a network that can solve this classification problem. Justify your choice of the number of nodes and the architecture. Draw the decision boundary that your network can find on the diagram. (4.0 Marks)

3.a. A Naïve Bayes sentiment classifier is given below, along with the training corpus:

Training Set	**Class Label(Sentiment)**
the movie has no plot	-ve
honestly pretty boring	-ve
pretty interesting movie	+ve
Test Set	
pretty enjoyable plot	?

Compute whether the sentence in the *test set* is of class positive or negative sentiment. **Demonstrate** a thorough step-by-step solution process. (6.0 Marks)

3.b. Give two differences between OOB Score and validation score? - 2M
What are the key hyperparameters of Gradient Boosting Model. - 3M
Write a note on Bootstrap aggregation. - 2M (7.0 Marks)

3.c. A simple implementation of the Naive Bayes Classifier for text classification is given below. This classifier assumes that the input features are binary word occurrences in documents. A code snippet for training the Naive Bayes Classifier using the provided training data is given. Explain what ‘dict_1’ and ‘dict_2’ are indicating.

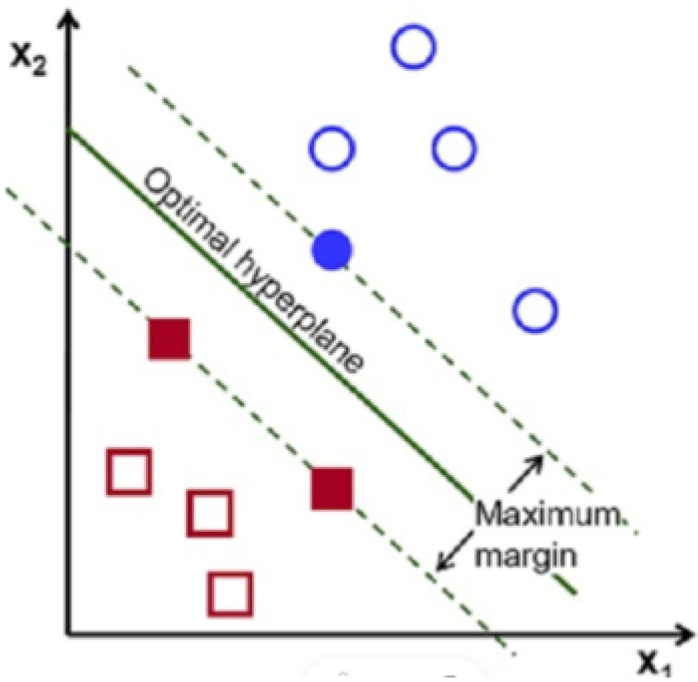
```
def fit(X, y):
    """
    Args:
      X (numpy.ndarray): The training data matrix where each row represents a document
          and each column represents the presence (1) or absence (0) of a word.
      y (numpy.ndarray): The corresponding labels for the training documents.

    Returns:
      tuple: A tuple containing two dictionaries: dict_1 and dict_2
    """
    classes = np.unique(y)
    dict_1 = {}
    dict_2 = {}
    total_documents = len(y)

    for c in classes:
      c_index = [index for index, value in enumerate(y) if value == c]
      X_c =  X[c_index]
      dict_1[c] = float(len(X_c)) / float(total_documents)
      dict_2[c] = (np.sum(X_c, axis = 0) + 1.0) / float(np.sum(X_c) + len(X[0]))

    return dict_1, dict_2
```

(4.0 Marks)

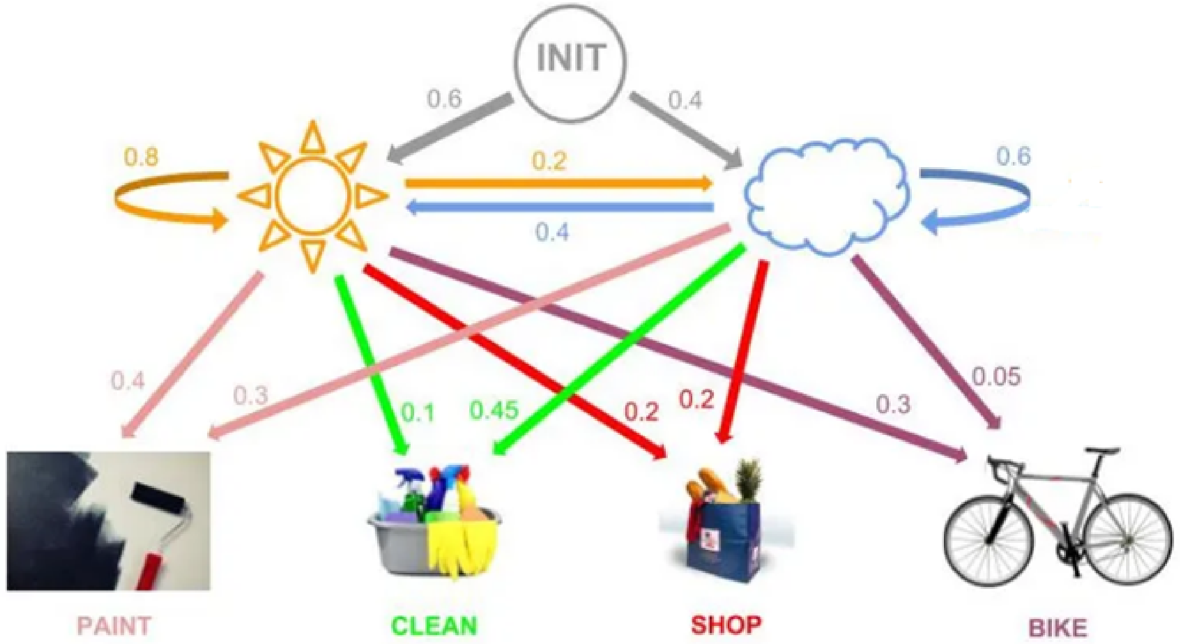
3.d. Using the Lagrange multiplier method, determine the dimensions of a rectangle (containing x as length and y as width) that will maximize its area, given that the perimeter is 20 meters. **Demonstrate** a thorough step-by-step solution process. (5.0 Marks)

3.e. Why SVM is an example of a large margin classifier? Explain using the given diagram.

(3.0 Marks)



4.a. Let's talk about Leena ; Leena is our imaginary friend and during the day she does either of these four things: *Painting, Cleaning the house, Biking, Shopping*. This is basically a list of what are called **observables**. For these observations we have two **hidden states** *Sunny or Rainy*. Here is a graph of potential HMM:



Looking at the graph **write** the tables for transition probabilities and emission probabilities.
Find the probabilities of the following.
i) Sequence "Cleaning Shopping" w.r.t "Sunny Rainy"
ii) Sequence "Party Cleaning" w.r.t. "Sunny Sunny"
iii) Sequence "Party Cleaning" w.r.t. "Rainy Sunny" (8.0 Marks)

4.b. Suppose a genetic algorithm uses chromosomes of the form x = abcdefgh with a fixed length of eight genes. Each gene can be any digit between 0 and 9.

Let the fitness of individual x be calculated as: f(x) = (a + b) − (c + d) + (e + f) − (g + h).

and let the initial population consist of four individuals with the following chromosomes:

x1 = 6 5 4 1 3 5 3 2
x2 = 8 7 1 2 6 6 0 1
x3 = 2 3 9 2 1 2 8 5
x4 = 4 1 8 5 2 0 9 4

1. Evaluate the fitness of each individual, showing all your workings, and arrange them in order with the fittest first and the least fit last
2. Perform the following crossover operations to obtain the **offspring**:
    1. Cross the fittest two individuals using one–point crossover at the middle point (*offsprings 1 and 2*)
    2. Cross the second and third fittest individuals using a two-point crossover (points b and f) for (*offspring 3 and 4)*
    3. Cross the first and third fittest individuals (ranked 1st and 3rd) using a uniform crossover (say, random points are a, d, and f) for (*offspring 5 and 6)*
3. Evaluate the fitness of the new population, showing all your workings. Has the overall fitness improved?

(8.0 Marks)

4.c. Explain Singular value decomposition.
What is Principal Component Analysis (PCA)? Which eigen value indicates the direction of largest variance? In what sense is the representation obtained from a projection onto the eigen directions corresponding the largest eigen values optimal for data reconstruction? (5.0 Marks)

4.d. What are some stopping criteria for K-means clustering? (4.0 Marks)